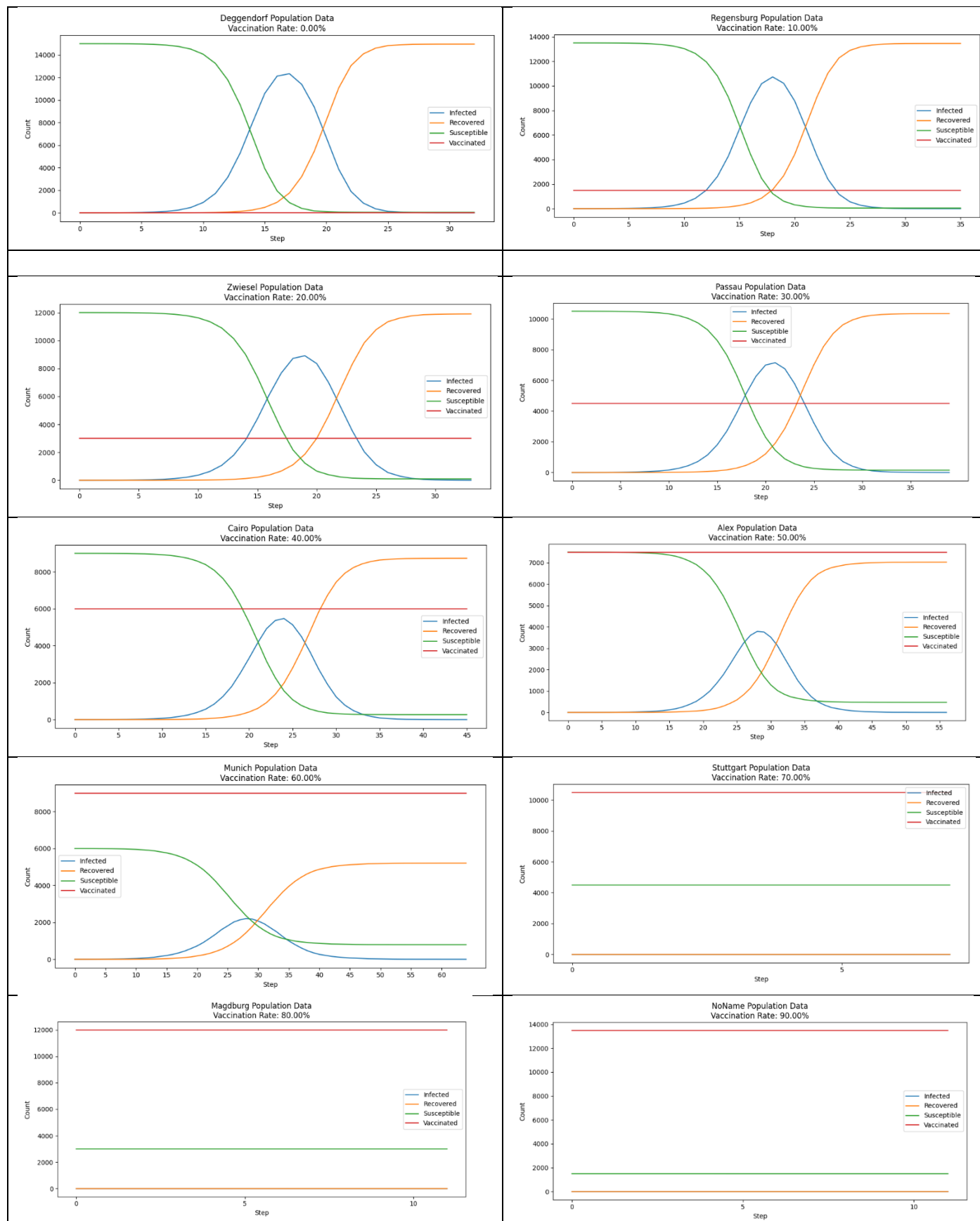


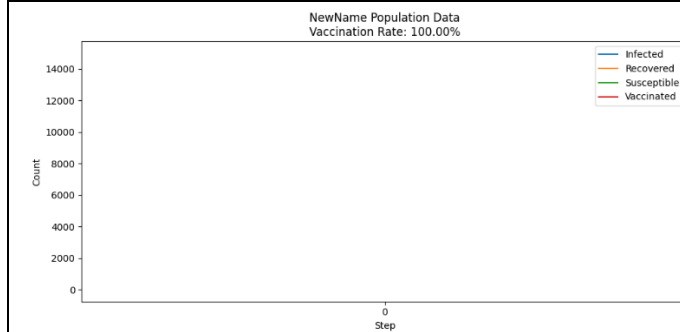
First experiment with the following parameters:

$D = 7$, $B = 0.1$ and $C = 6$. The basic reproduction number $R_0 = 4.2$.

The following table shows the infectious, recovered and susceptible people for each time step of a single population.

The table is sorted from right to left and from top to bottom in an ascending manner: Starting with 0% vaccination and reaching 100%. The final picture shows recovered people vs the vaccination rate at the end of each simulation.





All are 0 except for the vaccinated which is 15000 (For some reason the plot worked everywhere except for this part – which should have shown a dot at step 0).

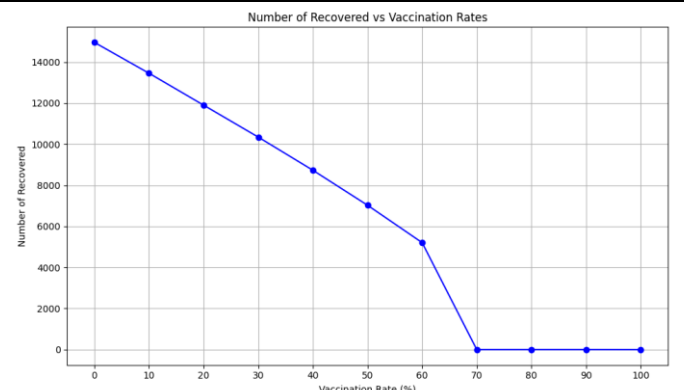


Table 1: Number of Infectious, recovered and susceptible people for each time step of a single simulation

Analysis:

From the herd immunity graph in Fig. 1, for $R_0 = 4.2$ the corresponding threshold is expected to be ≈ 0.76

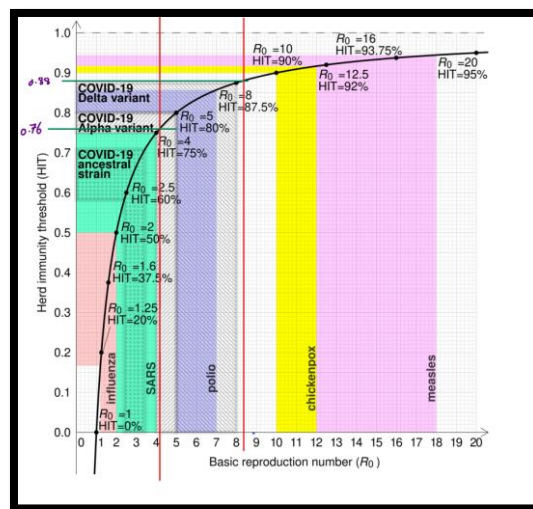
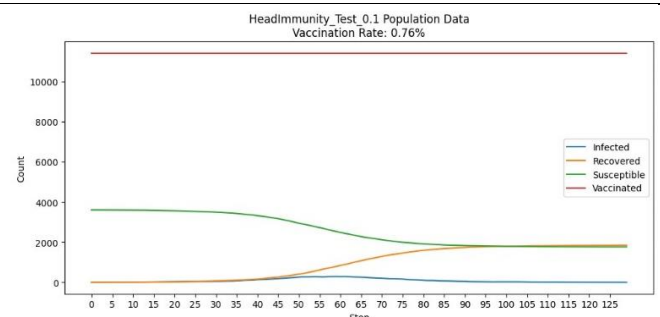


Figure 1: Herd immunity

After comparing the expected threshold with the previous figures in Table 1, the table indicates that the correct threshold is between 0.7 and 0.8 which is the expected range from Fig 1.

Simulating the exact value of *Vaccination Rate* = 0.76. The results match the expectation:

```
1 [global]
2 simulation_name = multiple
3 num_populations = 1
4 simulation_runs = 20
5
6
7 [disease] ; Global
8 name = "Corona" ; Name
9 duration = 7 ; Day
10 transmissibility = 0.1 ; Probability
11
12 ; For each population a second simulation
13 [population_1] ; Name
14 name = Deggendorf ; Name
15 size = 15000 ; Size
16 vaccination_rate = 0.76 ; Vaccination rate
17 patient_0 = true ; Patient 0
18
19 [population_2] ; Name
20 name = Regensburg ; Name
21 size = 15000 ; Size
22 vaccination_rate = 0.1 ; Vaccination rate
23 patient_0 = true ; Patient 0
24
25 [population_3] ; Name
26 name = Zwiesel ; Name
27 size = 15000 ; Size
28 vaccination_rate = 0.2 ; Vaccination rate
29 patient_0 = true ; Patient 0
30
31 [population_4] ; Name
32 name = Passau ; Name
33 size = 15000 ; Size
34 vaccination_rate = 0.3 ; Vaccination rate
35 patient_0 = true ; Patient 0
36
37 [population_5] ; Name
38 name = Munch ; Name
39 size = 15000 ; Size
40 vaccination_rate = 0.4 ; Vaccination rate
41 patient_0 = true ; Patient 0
42
43 [population_6] ; Name
44 name = Alex ; Name
45 size = 15000 ; Size
46 vaccination_rate = 0.5 ; Vaccination rate
47 patient_0 = true ; Patient 0
48
49 [population_7] ; Name
50 name = Munch ; Name
51 size = 15000 ; Size
52 vaccination_rate = 0.6 ; Vaccination rate
53 patient_0 = true ; Patient 0
```



The average graph after ≈ 20 simulations is quite similar to this one.

This simulated threshold lies between 0.7 and 0.8 as from the simulation output itself; shown in table 1.

After simulating the expected threshold from Fig. 1, results are approximately correct.

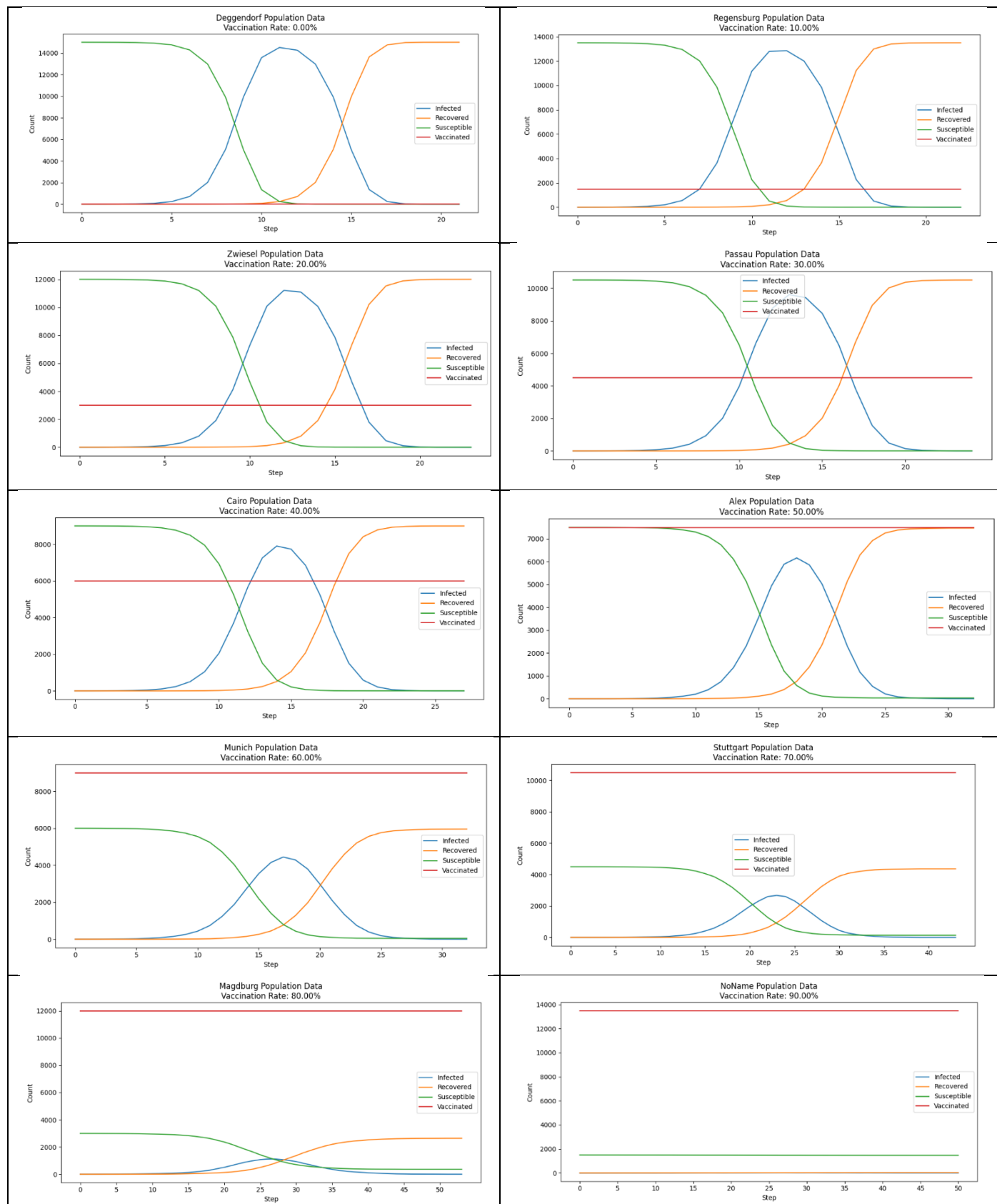
So, the simulated value is quite similar to the expected threshold behavior.

Second experiment with the following parameters:

$D = 7$, $B = 0.2$ and $C = 6$. The basic reproduction number $R_0 = 0.88$.

The following table shows the infectious, recovered and susceptible people for each time step of a single population.

The table is sorted from right to left and from top to bottom in an ascending manner: Starting with 0% vaccination and reaching 100%. The final picture shows recovered people vs the vaccination rate at the end of each simulation.



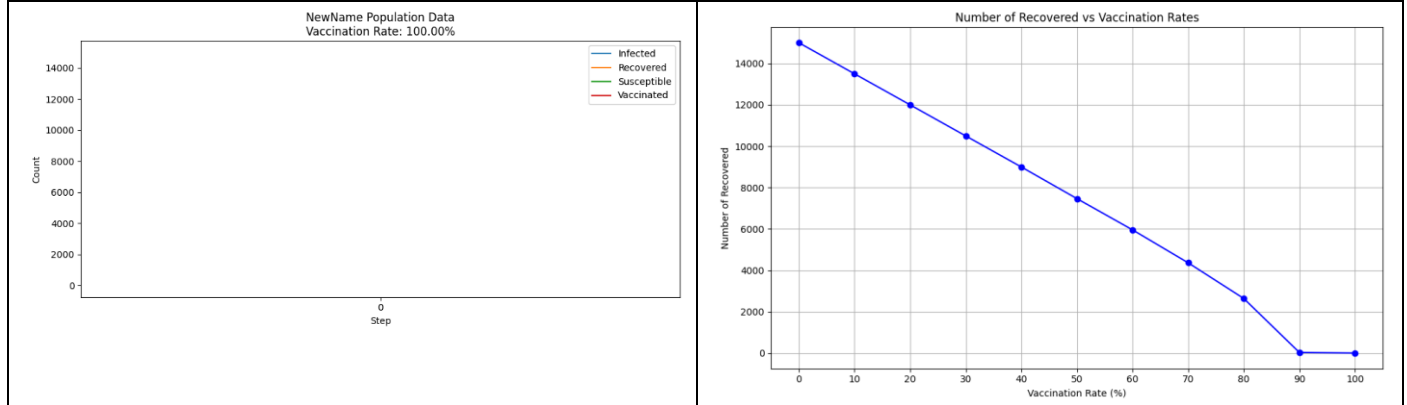


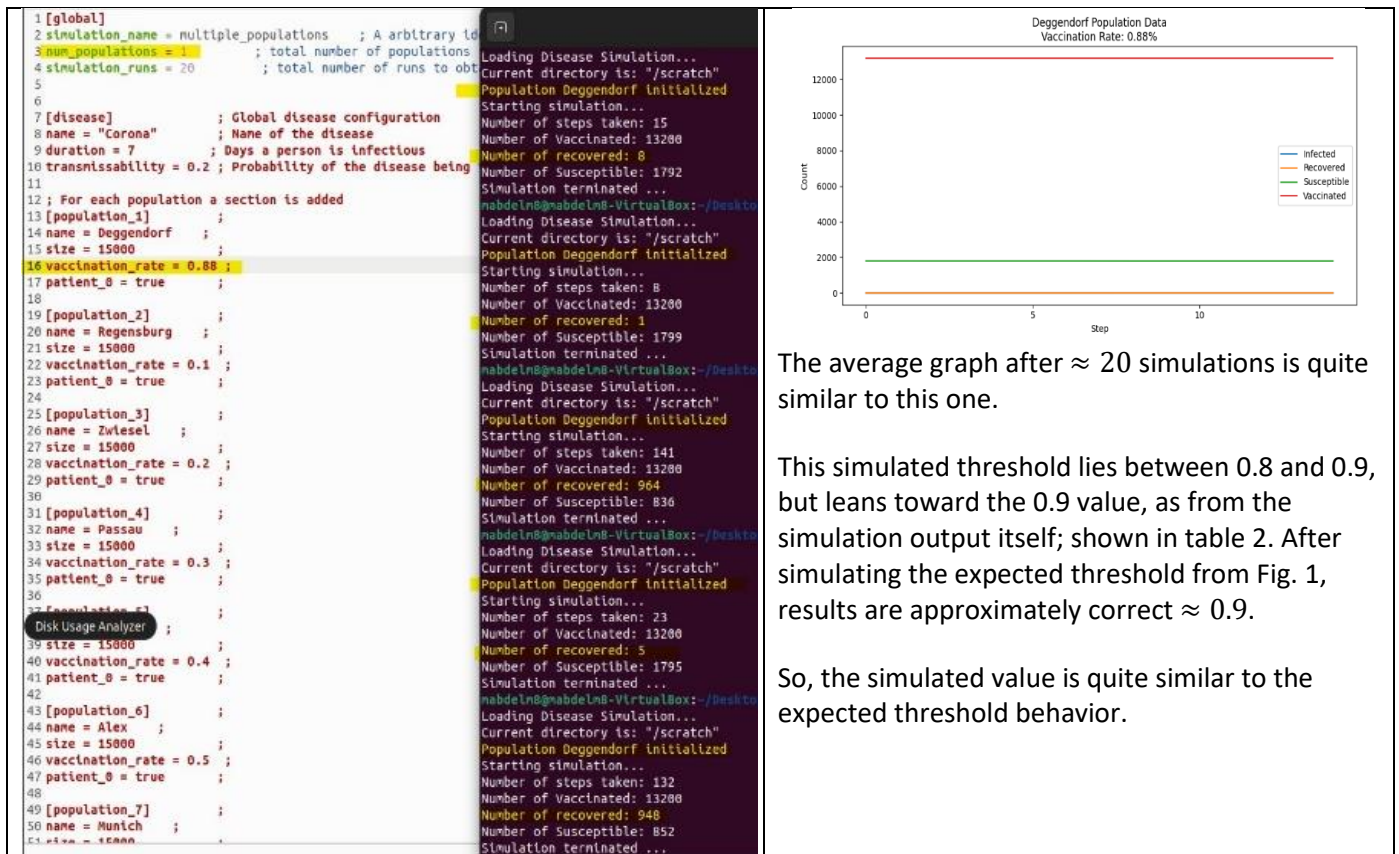
Table 2: Number of Infectious, recovered and susceptible people for each time step of a single simulation

Analysis:

From the herd immunity graph in Fig. 1, for $R_0 = 8.4$ the corresponding threshold is expected to be ≈ 0.88

After comparing the expected threshold with the previous figures in Table 2, the table indicates that the correct threshold is between 0.8 and 0.9, but close more to 0.9, which is the expected range from Fig 1.

Simulating the exact value of $Vaccination Rate = 0.88$. The results match the expectation:



The average graph after ≈ 20 simulations is quite similar to this one.

This simulated threshold lies between 0.8 and 0.9, but leans toward the 0.9 value, as from the simulation output itself; shown in table 2. After simulating the expected threshold from Fig. 1, results are approximately correct ≈ 0.9 .

So, the simulated value is quite similar to the expected threshold behavior.

Summary:

Simulation is working efficiently and the numbers are approximately equal to the expected threshold. The random generation process sometimes makes a spike in calculations, however the worst case results doesn't exceed 40% of number of simulations running; i.e. out of 100 runs, only 40 runs have a deviation in results.

For case study #1: The worst case for running the simulation was that 14% of the whole population got infected under the $Vaccination Rate = 0.76$ and the happy scenario reached 0.0133% of the population only got infected.

For case study #2: The worst case for running the simulation was that 7% of the whole population got infected under the $Vaccination Rate = 0.88$ and the happy scenario reached 0.0333% of the population only got infected.